

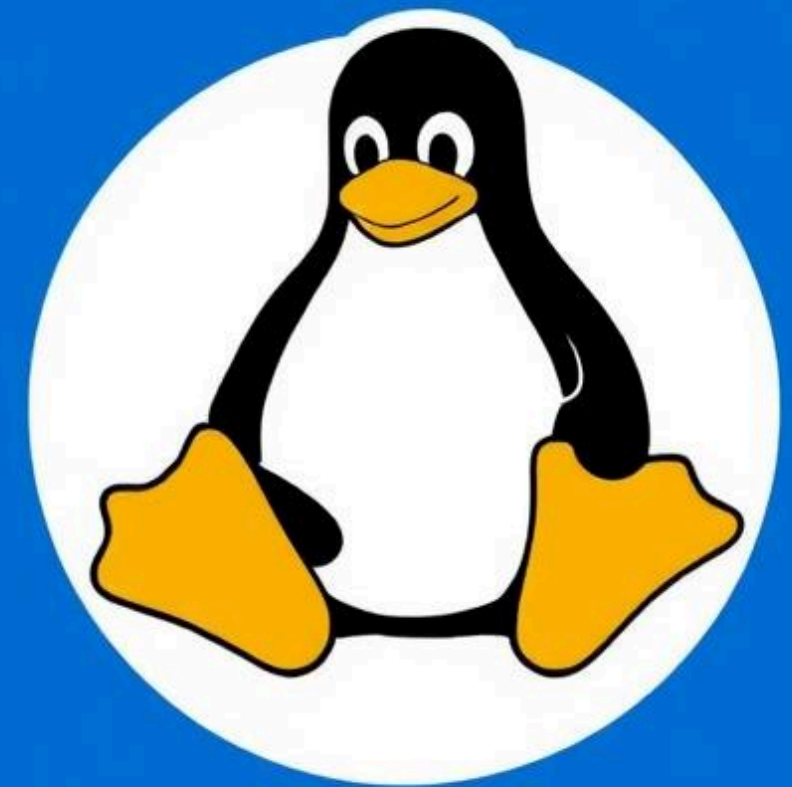
LINUX COMMANDS FOR CYBERSECURITY ANALYSTS



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LINUX COMMANDS FOR CYBERSECURITY ANALYSTS

Here is a curated list of 20 essential Linux commands for cybersecurity analysts, along with step-by-step examples for hands-on practice on Kali Linux. Each command includes a simple explanation and practical use cases.



IFCONFIG

Purpose: View and configure network interfaces.

- Example:
- ifconfig

Displays active network interfaces, IP addresses, netmask, and MAC addresses.

```
kali@kali: ~  
File Actions Edit View Help  
(kali@kali)-[~]  
$ ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 192.168.64.5 netmask 255.255.255.0 broadcast 192.168.64.255  
    inet6 fd99:7f62:7090:d13e:1058:84ff:feb6:ac09 prefixlen 64 scopeid  
0x0<global>  
    inet6 fd99:7f62:7090:d13e:253a:dce1:3e8e:b655 prefixlen 64 scopeid  
0x0<global>  
    inet6 fe80::1058:84ff:feb6:ac09 prefixlen 64 scopeid 0x20<link>  
    ether 12:58:84:b6:ac:09 txqueuelen 1000 (Ethernet)  
    RX packets 33 bytes 22144 (21.6 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 27 bytes 4570 (4.4 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0x10<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 8 bytes 480 (480.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 8 bytes 480 (480.0 B)
```

PING

Purpose: Test connectivity to a remote host.

- Example:
- `ping google.com`

Sends ICMP packets to check if the host is reachable.

```
(kali@kali)-[~]  
$ ping google.com  
PING google.com (142.250.191.142) 56(84) bytes of data.  
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=1 ttl=113 time  
=16.0 ms  
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=2 ttl=113 time  
=13.8 ms  
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=3 ttl=113 time  
=14.1 ms  
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=4 ttl=113 time  
=13.5 ms  
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=5 ttl=113 time  
=12.7 ms  
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=6 ttl=113 time  
=13.6 ms  
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=7 ttl=113 time  
=12.2 ms  
64 bytes from ord38s29-in-f14.1e100.net (142.250.191.142): icmp_seq=8 ttl=113 time  
=12.6 ms
```


NETSTAT

Purpose: Display network connections and listening ports.

- Example:
- `netstat -tuln`

Shows active TCP/UDP connections and listening ports.

```
(kali@kali)-[~]
$ netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State

```



```
(kali@kali)-[~]
$ netstat ?
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp      0      0 192.168.64.5:bootpc     192.168.64.1:bootps     ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags   Type       State         I-Node  Path
unix  3      [ ]     STREAM    CONNECTED    9450
unix  3      [ ]     STREAM    CONNECTED    10446    @/tmp/.X11-unix/X0
unix  3      [ ]     STREAM    CONNECTED    9302
unix  3      [ ]     DGRAM     CONNECTED    7678
unix  3      [ ]     STREAM    CONNECTED    9275     /run/user/1000/bus
unix  3      [ ]     DGRAM     CONNECTED    6602
unix  3      [ ]     STREAM    CONNECTED    9368
unix  3      [ ]     STREAM    CONNECTED    9444     /run/user/1000/bus
unix  3      [ ]     STREAM    CONNECTED    9306
```

NMAP

Purpose: Perform network scanning and port enumeration.

- Example:
- `nmap -sS -p 1-1000 <target>`

Conducts a stealth scan on ports 1-1000 of the target.

```
(kali㉿kali)-[~]  
└─$ nmap -sS -p 1-1000 infosecclabs.io  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-27 00:50 CDT  
Nmap scan report for infosecclabs.io (172.67.206.246)  
Host is up (0.017s latency).  
Other addresses for infosecclabs.io (not scanned): 104.21.69.94 2606:4700:3035::ac4  
3:cef6 2606:4700:3030::6815:455e  
Not shown: 998 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
80/tcp    open  http  
443/tcp   open  https  
  
Nmap done: 1 IP address (1 host up) scanned in 5.49 seconds
```

Please do not try nmap to any website without getting approval of the owner. This is for education purpose only.

TCPDUMP

Purpose: Capture and analyze network traffic.

- Example:
- `sudo tcpdump -i eth0`

Captures packets on the eth0 interface.

```
(kali㉿kali)-[~]  
$ sudo tcpdump -i eth0  
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode  
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes  
01:18:13.626803 IP6 fe80::a078:17ff:fe7a:d764 > ip6-allnodes: ICMP6, router advert  
isement, length 88  
01:18:13.639180 IP6 fe80::1058:84ff:feb6:ac09 > ff02::16: HBH ICMP6, multicast lis  
tener report v2, 2 group record(s), length 48  
01:18:13.686055 IP6 fe80::1058:84ff:feb6:ac09 > ff02::16: HBH ICMP6, multicast lis  
tener report v2, 2 group record(s), length 48  
01:18:13.691461 IP 192.168.64.5.40061 > 192.168.64.1.domain: 28011+ PTR? 4.6.7.d.a  
.7.e.f.f.f.7.1.8.7.0.a.0.0.0.0.0.0.0.0.0.0.0.0.8.e.f.ip6.arpa. (90)  
01:18:13.697538 ARP, Request who-has 192.168.64.5 (12:58:84:b6:ac:09 (oui Unknown)  
) tell 192.168.64.1, length 28  
01:18:13.697616 ARP, Reply 192.168.64.5 is-at 12:58:84:b6:ac:09 (oui Unknown), len  
gth 28  
01:18:13.697538 IP 192.168.64.1.domain > 192.168.64.5.40061: 28011 NXDomain* 0/0/0  
(90)  
01:18:13.701092 IP 192.168.64.5.43800 > 192.168.64.1.domain: 48494+ PTR? 9.0.c.a.6  
.b.e.f.f.f.4.8.8.5.0.1.0.0.0.0.0.0.0.0.0.0.0.0.8.e.f.ip6.arpa. (90)  
01:18:13.707134 IP 192.168.64.1.domain > 192.168.64.5.43800: 48494 NXDomain* 0/0/0
```

WHOIS

Purpose: Retrieve WHOIS information for domains.

- Example:
- `whois example.com`

Displays registration details of the domain.

```
(kali@kali)-[~]  
$ whois google.com  
Domain Name: GOOGLE.COM  
Registry Domain ID: 2138514_DOMAIN_COM-VRSN  
Registrar WHOIS Server: whois.markmonitor.com  
Registrar URL: http://www.markmonitor.com  
Updated Date: 2019-09-09T15:39:04Z  
Creation Date: 1997-09-15T04:00:00Z  
Registry Expiry Date: 2028-09-14T04:00:00Z  
Registrar: MarkMonitor Inc.  
Registrar IANA ID: 292  
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com  
Registrar Abuse Contact Phone: +1.2086851750  
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited  
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited  
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited  
Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
```

DIG

Purpose: Perform DNS queries.

- Example:
- dig google.com

Retrieves DNS records for the domain.

```
(kali@kali)-[~]
$ dig google.com

; <<>> DiG 9.20.4-4-Debian <<>> google.com
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 7295
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags::; udp: 4096
;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                 289     IN      A      142.250.191.142

;; Query time: 7 msec
;; SERVER: 192.168.64.1#53(192.168.64.1) (UDP)
;; WHEN: Thu Mar 27 01:24:25 CDT 2025
;; MSG SIZE  rcvd: 55
```


TRACEROUTE

Purpose: Trace the route packets take to a destination.

- Example:
- `tracert google.com`

Shows the path packets travel to reach the target.

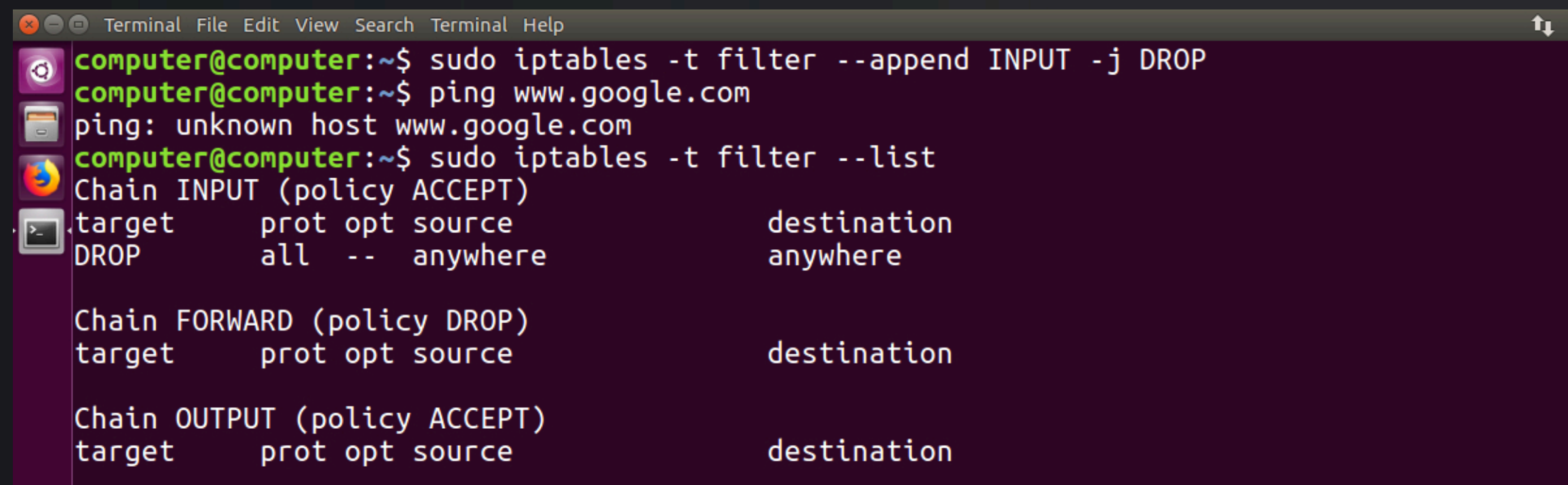
```
(kali@kali)-[~]  
$ tracert google.com  
tracert to google.com (142.250.191.142), 30 hops max, 60 byte packets  
 1 * 192.168.64.1 (192.168.64.1) 0.339 ms 0.327 ms  
 2 192.168.40.1 (192.168.40.1) 6.227 ms 6.216 ms 6.206 ms  
 3 10.255.235.13 (10.255.235.13) 12.425 ms 12.416 ms 11.214 ms  
 4 216.169.31.68 (216.169.31.68) 9.229 ms 216.169.31.84 (216.169.31.84) 8.883 m  
s 8.679 ms  
 5 216.169.31.40 (216.169.31.40) 8.606 ms 216.169.31.36 (216.169.31.36) 6.522 m  
s 8.419 ms  
 6 e0-32.core1.stl2.he.net (184.105.12.197) 10.375 ms 10.239 ms 10.210 ms  
 7 port-channel11.core1.stl1.he.net (184.104.193.168) 7.252 ms 7.732 ms 8.014  
ms  
 8 * port-channel32.core3.chi1.he.net (184.104.188.170) 13.686 ms *  
 9 eqix-ch-200g-1.google.com (208.115.136.21) 13.665 ms 15.459 ms 13.645 ms  
10 * 209.85.246.95 (209.85.246.95) 15.244 ms 15.234 ms  
11 142.251.60.7 (142.251.60.7) 13.420 ms 142.251.60.214 (142.251.60.214) 15.110  
ms 142.251.60.200 (142.251.60.200) 15.079 ms  
12 216.239.54.16 (216.239.54.16) 23.914 ms ord38s29-in-f14.1e100.net (142.250.19  
1.142) 13.251 ms 13.191 ms
```

IPTABLES

Purpose: Configure firewall rules.

- Example:
- `sudo iptables -t filter --append INPUT -j DROP`

This command drops all the traffic coming on any port.

A terminal window with a dark background and light text. The window title is "Terminal" with standard window controls. The prompt is "computer@computer:~\$". The first command is "sudo iptables -t filter --append INPUT -j DROP". The second command is "ping www.google.com", which returns "ping: unknown host www.google.com". The third command is "sudo iptables -t filter --list", which lists the current iptables rules. The output shows three chains: INPUT (policy ACCEPT), FORWARD (policy DROP), and OUTPUT (policy ACCEPT). Each chain has a table with columns: target, prot, opt, source, and destination. The INPUT chain has a rule with target DROP, protocol all, and destination anywhere. The FORWARD and OUTPUT chains have no rules listed.

```
computer@computer:~$ sudo iptables -t filter --append INPUT -j DROP
computer@computer:~$ ping www.google.com
ping: unknown host www.google.com
computer@computer:~$ sudo iptables -t filter --list
Chain INPUT (policy ACCEPT)
target     prot opt source                destination
DROP      all  --  anywhere               anywhere

Chain FORWARD (policy DROP)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination
```

Source: <https://www.geeksforgeeks.org/iptables-command-in-linux-with-examples/>

SSH

Purpose: Securely connect to remote systems.

- Example:
- `ssh user@192.168.1.10`

Connects to a remote server using SSH.

```
(kali@kali)-[~]  
$ ssh ebayar@192.168.40.203  
The authenticity of host '192.168.40.203 (192.168.40.203)' can't be established.  
ED25519 key fingerprint is SHA256:FkMzbFR6jXXF92kLibAWS9NhSe1ZChh7GBaeBnz+818.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? y  
Please type 'yes', 'no' or the fingerprint: yes  
Warning: Permanently added '192.168.40.203' (ED25519) to the list of known hosts.  
(ebayar@192.168.40.203) Password:  
Last login: Sun Mar 23 22:33:09 2025  
ebayar@MacBookPro ~ %
```


WGET

Purpose: Download files from the internet.

- Example:
- wget
<https://example.com/file.txt>

Downloads the specified file.

```
sara@phoenixnap:~$ wget -O PuppetServer https://apt.puppetlabs.com/puppet8-release-jammy
.deb
--2024-02-13 15:03:26-- https://apt.puppetlabs.com/puppet8-release-jammy.deb
Resolving apt.puppetlabs.com (apt.puppetlabs.com)... 13.32.110.100, 13.32.110.3, 13.32.1
10.9, ...
Connecting to apt.puppetlabs.com (apt.puppetlabs.com)|13.32.110.100|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 11316 (11K) [application/x-debian-package]
Saving to: 'PuppetServer'

PuppetServer          100%[=====>] 11.05K  --.-KB/s    in 0s

2024-02-13 15:03:27 (76.5 MB/s) - 'PuppetServer' saved [11316/11316]
```

Source: <https://phoenixnap.com/kb/wget-command-with-examples>

CURL

Purpose: Transfer data from or to a server.

- Example:
- `curl https://www.infoseclabs.io`

Retrieves the HTML content of the specified URL and display it in the terminal.

```
(kali㉿kali)-[~]  
$ curl www.infoseclabs.io  
<html>  
<head><title>301 Moved Permanently</title></head>  
<body>  
<center><h1>301 Moved Permanently</h1></center>  
<hr><center>cloudflare</center>  
</body>  
</html>
```

AIRCRAK-NG

Purpose: Crack Wi-Fi passwords for security testing.

- Example:
- aircrack-ng capture_file.cap -w wordlist.txt

Attempts to crack encryption using a wordlist.

```
(kali@kali)-[~]  
$ aircrack-ng  
  
Aircrack-ng 1.7 - (C) 2006-2022 Thomas d'Otreppe  
https://www.aircrack-ng.org  
  
usage: aircrack-ng [options] <input file(s)>  
  
Common options:  
  
-a <amode> : force attack mode (1/WEP, 2/WPA-PSK)  
-e <essid> : target selection: network identifier  
-b <bssid> : target selection: access point's MAC  
-p <nbcpu> : # of CPU to use (default: all CPUs)  
-q          : enable quiet mode (no status output)  
-C <macs>  : merge the given APs to a virtual one  
-l <file>  : write key to file. Overwrites file.  
  
Static WEP cracking options:  
  
-c          : search alphanumeric characters only  
-t          : search binary coded decimal chr only  
-h          : search the numeric key for Fritz!BOX  
-d <mask>   : use masking of the key (A1:XX:CF:YY)  
-m <maddr>  : MAC address to filter usable packets  
-n <nbits>  : WEP key length : 64/128/152/256/512
```


METASPLOIT

Purpose: Exploit vulnerabilities in systems.

- Example:
- msfconsole

Launches Metasploit framework for penetration testing.

```
(kali㉿kali)-[~]  
$ msfconsole  
Metasploit tip: You can pivot connections over sessions started with the  
ssh_login modules  
  
Unable to handle kernel NULL pointer dereference at virtual address 0xd34db33f  
EFLAGS: 00010046  
eax: 00000001 ebx: f77c8c00 ecx: 00000000 edx: f77f0001  
esi: 803bf014 edi: 8023c755 ebp: 80237f84 esp: 80237f60  
ds: 0018  es: 0018  ss: 0018  
Process Swapper (Pid: 0, process nr: 0, stackpage=80377000)  
  
Stack: 90909090909090909090909090909090  
90909090909090909090909090909090  
90909090.90909090.90909090  
90909090.90909090.90909090  
90909090.90909090.09090900  
90909090.90909090.09090900  
.....  
cccccccccccccccccccccccccccccccc  
cccccccccccccccccccccccccccccccc  
cccccccccc.....  
cccccccccccccccccccccccccccccccc  
cccccccccccccccccccccccccccccccc
```

HYDRA

Purpose: Perform brute-force login attacks.

- Example:
- `hydra -l admin -P passwords.txt <target> ssh`

Attempts SSH login using a password list.

```
(kali㉿kali)-[~]  
$ hydra -help  
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in mili  
tary or secret service organizations, or for illegal purposes (this is non-binding  
, these ** ignore laws and ethics anyway).  
  
Syntax: hydra [[[-l LOGIN|-L FILE] [-p PASS|-P FILE]] | [-C FILE]] [-e nsr] [-o FI  
LE] [-t TASKS] [-M FILE [-T TASKS]] [-w TIME] [-W TIME] [-f] [-s PORT] [-x MIN:MAX  
:CHARSET] [-c TIME] [-ISOuvVd46] [-m MODULE_OPT] [service://server[:PORT][/OPT]]  
  
Options:  
-R          restore a previous aborted/crashed session  
-I          ignore an existing restore file (don't wait 10 seconds)  
-S          perform an SSL connect  
-s PORT     if the service is on a different default port, define it here  
-l LOGIN or -L FILE login with LOGIN name, or load several logins from FILE  
-p PASS or -P FILE try password PASS, or load several passwords from FILE  
-x MIN:MAX:CHARSET password bruteforce generation, type "-x -h" to get help  
-y          disable use of symbols in bruteforce, see above  
-r          use a non-random shuffling method for option -x  
-e nsr     try "n" null password, "s" login as pass and/or "r" reversed login  
-u         loop around users, not passwords (effective! implied with -x)  
-C FILE    colon separated "login:pass" format, instead of -L/-P options  
-M FILE    list of servers to attack, one entry per line, ':' to specify port  
-o FILE    write found login/password pairs to FILE instead of stdout  
-b FORMAT  specify the format for the -o FILE: text(default), json, jsonv1
```

JOHN

Purpose: Crack hashed passwords.

- Example:
- `john --wordlist=wordlist.txt hashfile.txt`

Cracks hashes using a wordlist.

```
(kali@kali)-[~]  
$ john  
Created directory: /home/kali/.john  
John the Ripper 1.9.0-jumbo-1+bleeding-aec1328d6c 2021-11-02 10:45:52 +0100  
linux-gnu 64-bit aarch64 ASIMD AC]  
Copyright (c) 1996-2021 by Solar Designer and others  
Homepage: https://www.openwall.com/john/  
  
Usage: john [OPTIONS] [PASSWORD-FILES]  
  
Use --help to list all available options.
```

FIND

Purpose: Locate files and directories.

- Example:
- `find /home -name "*.txt"`

Searches for .txt files in /home.

```
(kali@kali)-[~]  
$ find /home -name "*.txt"  
/home/kali/.mozilla/firefox/oo7f0upg.default-esr/serviceworker.txt  
/home/kali/.mozilla/firefox/oo7f0upg.default-esr/pkcs11.txt
```

```
(kali@kali)-[~]  
$
```


CHMOD

Purpose: Change file permissions.

- Example:
- `chmod +x script.sh`

Makes the file executable.

```
(kali@kali)-[~]
$ chmod --help
Usage: chmod [OPTION] ... MODE[,MODE] ... FILE ...
       or:  chmod [OPTION] ... OCTAL-MODE FILE ...
       or:  chmod [OPTION] ... --reference=RFILE FILE ...
Change the mode of each FILE to MODE.
With --reference, change the mode of each FILE to that of RFILE.

-c, --changes           like verbose but report only when a change is made
-f, --silent, --quiet  suppress most error messages
-v, --verbose          output a diagnostic for every file processed
                       --dereference  affect the referent of each symbolic link,
                                   rather than the symbolic link itself
-h, --no-dereference   affect each symbolic link, rather than the referent
--no-preserve-root     do not treat '/' specially (the default)
--preserve-root        fail to operate recursively on '/'
--reference=RFILE      use RFILE's mode instead of specifying MODE values.
```

PS

Purpose: View active processes on the system.

- Example:
- `ps aux | grep apache2`

Lists processes related to Apache web server.

```
(kali㉿kali)-[~]  
$ ps aux | grep apache2  
kali      51232  0.0  0.0   6236  1864 pts/0    S+   02:16   0:00 grep --color=au  
to apache2  
  
(kali㉿kali)-[~]  
$
```

HISTORY

Purpose: View previously executed commands.

- Example:
- `history | grep nmap`

Displays all past commands containing "nmap."

```
(kali㉿kali)-[~]  
$ history | grep nmap  
12  nmap -sS -p 1-1000 infosecclabs.io  
  
(kali㉿kali)-[~]  
$
```


OUR TEAM

Let's meet our team members
who are professional and have
rich experiences.



HALIL BARIS
FOUNDER



ERKAN BAYAR
DIRECTOR OF EDUCATION

THANK YOU

FOR WATCHING